

A REAL-TIME PROJECT REPORT ON

**RFID AND FINGERPRINT BASED EXAMINATION ROOM
IDENTIFICATION**

**A PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENT FOR THE AWARD OF THE DEGREE OF**

**BACHELOR OF TECHNOLOGY
IN**

Electronics & Communication Engineering

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

**CMR TECHNICAL CAMPUSUGC
AUTONOMOUS**

Kandlakoya (V), Medchal Road, Hyderabad-501401 (2023 2024)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

This is to certify that the dissertation work entitled **“RFID BASED EXAMINATION ROOM IDENTIFICATION”** is being submitted by **G.SUSMITHA, K.DHANUNJAY, J. DHANUSH and T.KIRAN CHAITHANYA** bearing **Roll Numbers 227R1A0481, 227R1A0485, 227R1A0469, and 227R1A04B9** respectively, in partial fulfillment for the degree of **Bachelor of Technology** in **“ELECTRONICS AND COMMUNICATION ENGINEERING”** during the academic year 2020–2021. Certified further, to the best of our knowledge that the work reported is not a part of any other project on the basis of which a degree or an award has been given on an earlier occasion or any other candidate. The results have been verified and found to be satisfactory.

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ABSTRACT

The goal of this project is to provide guidance to students at examination centres. Many students are having difficulty finding rooms. Most students are worry while going to the examination, and they even feel tense while looking for their exam location. In these instances, our project is beneficial. As a hall ticket, each student will be given an RFID card. That will automatically display the room number of the student. Our project will primarily save time spent searching for a room in examination centres. KEY WORDS: Arduino, Examination room guidance, RFID system, Fingerprint module, Lcd display.

The advent of fast growing technologies makes users to have high security systems with electronic identification options. These identification technologies include Exam Hall Authentication user ID and password based systems, and so on. But, unfortunately these are unsecure due to hacker attacks and forgotten passwords. In spite of all these shortcomings and malfunctions these systems are still prevailing; however, the biometric or fingerprint authentication based identification is the efficient and reliable solution for stringent security.

Keywords- Fingerprint module, LCD Display, Microcontroller

Recognition of person on the basis of biometric future is an emerging phenomenon in our society. It Might involve validating personal identity and also other identity. In examination process we can use the biometric authentication for reducing impersonation. It is a security process that we release on the unique biological characteristics of an individual to verify that whether he/she valid for this examination hall.

The System is designed to pass only uses by verifying their finger print module was designed to scan the finger print, ID number and also other personal details which were properly saved into database of the system and confirm the eligibility of candidate for examination.

To create a system that is capable of tracking impersonators in the examination system using the methodology of finger print biometrics. To reduce rate of corruption in the educational sector and increase the rate of self-confidence on students. Keywords: Authentication, Finger print, Impersonation, ID number, other persona

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CHAPTER 1:

INTRODUCTION

RFID (Radio Frequency Identification) technology is a new technology that is being used in a various application. It is a member of the family of Automatic Identification and Data Capture (AIDC) technologies that provides a quick and dependable means of identifying objects. RFID is made up of two major components.

The Interrogator (RFID Reader) sends and receives the signal, as well as the Transponder (tag) attached to the object. RFID tags are "interrogated" by an RFID reader in an RFID system. The tag reader sends out a radio frequency "interrogation" in order to communicate with the tags. The reader also has a receiver that receives and decodes a response signal from the tags.

RFID (Radio Frequency Identification) technology is a new technology that is being used in a various application. RFID is made up of two major components. The Interrogator (RFID Reader) sends and receives the signal, as well as the Transponder (tag) attached to the object.

Improvements in RFID technology:

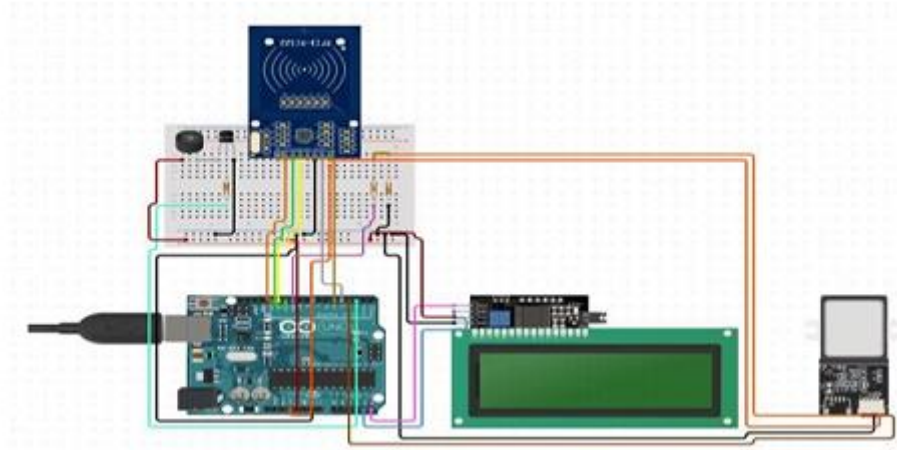
memory capacities, reading ranges, and processing speeds continue to improve. The integrated circuit in an RF tag will never be as cost effective as a bar code label

The tag's data content is reflected in the tag's response signal. A passive "backscatter" signal is used to generate the response signal. An RFID tag is made up of a tiny microchip and an antenna. RFID has numerous applications on its own, but when combined with a microcontroller, the possibilities expand even further.

Improvements in RFID technology: memory capacities, reading ranges, and processing speeds continue to improve. The integrated circuit in an RF tag will never be as cost effective as a bar code label, notwithstanding inevitable reductions in raw materials and economies of scale. RFID, on the other hand, will continue to expand in its established niches where bar code or other optical technologies are ineffective. If some level of standardisation is achieved, the market will very probably develop tremendously if RFID hardware from various companies can be used interchangeably

1.2 Project objective

The aim of the project is to utilize RFID system in a different and developmental side. We all know that RFID is used in each and every industrial and manufacturing field. Our major focus is to utilize this tool towards the residents and commercial users especially the old and disabled ones.



This form of system is called RFID based attendance system which focuses on making it possible for the elderly and disabled to remain at school, safe and comfortable. Attendance system is becoming a viable option for the elderly and disabled who would prefer to stay in the comfort of their homes rather than move to a healthcare facility.

1.3 Project scope

In addition to the present system, a facial recognition system can be incorporated for increased safety. More capabilities are being added, such as the ability to keep track of a student's details, such as fees due, library transactions, and attendance. The notion is not only useful to students but also to corporations, depending on how well it is implemented, as seen by the seeds that have sprouted into a variety of viable projects.



CHAPTER 2:

2.1 LITERATURE SURVEY

[1] C VijayaBhaskar1,P. RuchitaThe system is designed to pass only uses by verifying their fingerprint module The ID and the other personal details which where propely saved in database of the system and confirm the eligibility of candidate for examination The tag identification by our nacked eye.

[2] Z. yongqang.A wireless fingerprint module based on the attendance system which uses the fingerprint module AR400 RFID and handled MC 9000-GRFID reduces human effort and error

[3] Mr.N.JyothiSwaroop Mr. P.NagarjunaandMr. Ch. Seshu kumar. By combining both RFID and microcontroller generates a project with wider boundaries

and effective solutions. More capabilities are being added, such as the ability to keep track of a student's details, such as fees due.

[4] S. Martin', R. Gil', J. Bravo. This includes reducing waiting times for class access and real-time identification of students entering and exiting the examination room. The coverage range of RFID is limited which is about 3 meters.

[5] K. Vandana, K. Anil Kumar. For the existing system we can also add figure print authentication and face recognition system can also be implemented for the purpose of better safety. Reduces human effort and error

[6] Mr. D. S. Chaware¹, Mr. V. M. Pund. By integrating RFID Technology, we could successfully develop an embedded system for examination room guidance. All RFID tags within range can be detected instantly and matched with information in your database.

[7] A. N. Nambiar **A Supply Chain Perspective of RFID Systems.** RFID systems facilitate automatic and remote identification of components Tag identification that is not visible to the naked eye.

[8] *RKAR. Kariapper, MS. *Suhail Razeeth. **A RFID BASED (IoT) AUTOMATIC ATTENDANCE SYSTEM: A SURVEY ANALYSIS** This attendance system software has been developed using VB.net and database (Microsoft Access).

CHAPTER 3:

3.1 EXISTING METHOD

This project proposes a jumbling system-based examination solution. It is possible that students will have difficulty locating their respective rooms as a result of this. Using RFID technology, this system assists in locating exam halls and seats. Each student is given an RFID tag. A valid candidate will be able to easily locate his examination venue using RFID Technology. In existing systems, the researchers created a concept of a secure and adaptable embedded reader system. Another existing system focuses on supply chain management and makes use of RFID. The usage of RFID in a semiconductor technology to handle inventory transaction concerns is another topic under discussion

This system lowers costs and eliminates human error. Automated attendance management is a system in which they used both an electronic and a mobile platform, respectively, with a stationary matrix AR 400 RFID reader and a handheld MC 9000-G RFID reader. Zhang Young created a wireless fingerprint-based attendance system that uses finger prints to record and retrieve attendance data. The RFID reader will interrogate the student's swiped tag and obtain the information contained within it. This will be passed on to the controller. A unique passcode is provided for added security, and the student must enter it using the keypad. If both the id and passcode are correct, the candidate's room number will

be displayed on the LCD panel connected to the controller, marking him as an authorised entry.

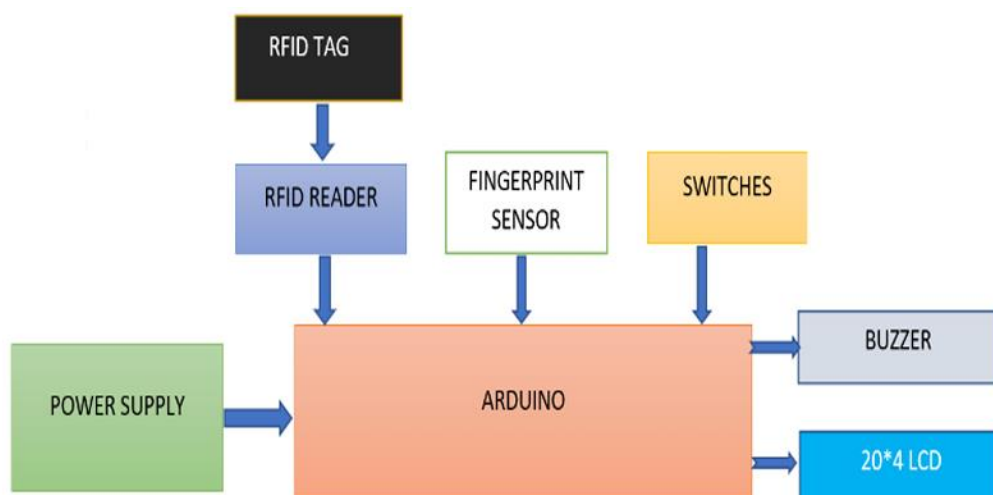
When a student enters the Examination Centre, he must keep his RFID TAG attached to the reader, which is connected to our microcontroller-based embedded board. At that point, the reader will determine whether or not the student's information is correct, and it will then decide whether or not to display the student's room number on the LCD display. When a student enters the Examination Centre, he must keep his RFID TAG attached to the reader, which is connected to our microcontroller-based embedded board. At that point, the reader will determine whether or not the student's information is correct, and it will then decide whether or not to display the student's room number on the LCD display.

Proposed system:

Using RFID technology, this system assists in locating exam halls and seats. Each student is given an RFID tag. The block diagram is depicted in below figure1.

The proposed system consists of Arduino NANO, RFID Reader, RFID Tags, LCD, Buzzer, Fingerprint sensor. Arduino NANO: The Arduino Nano is a compact, breadboard-friendly board based on the ATmega328 (Arduino Nano 3.0) or ATmega168 microcontroller (Arduino Nano 2.0). (Arduino Nano version 2.x) It is similar to the Arduino in terms of capabilities. The Arduino NANO is depicted in Figure 2.

Block diagram:



Rfid and fingerprint based examination room identification block diagram

Figure (1)

Problem statement

In the context of examination centers, especially those implementing a jumbling system, students often face challenges in locating their examination rooms. This can lead to anxiety and confusion, potentially impacting their performance. The goal is to develop an **Examination Room Guidance System** using RFID (Radio Frequency Identification) technology to streamline the process of guiding students to their respective rooms efficiently.

Methodology

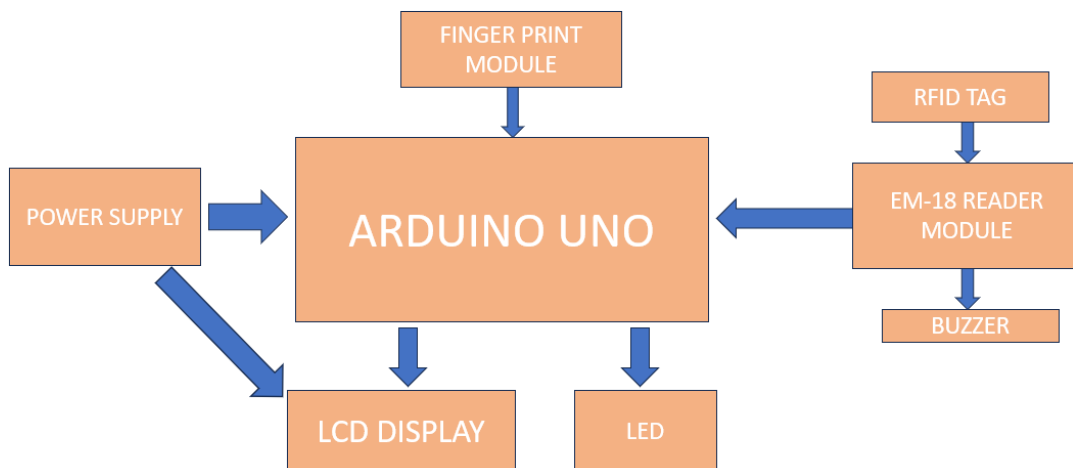


Figure (2)

The methodology for an RFID-based examination room identification system typically involves several key components and steps. Below is a detailed outline of the methodology:

- **Fingerprint Module to Arduino:** Connect the fingerprint module's TX and RX pins to the corresponding RX and TX pins on the Arduino (using a software serial port).
- **EM-18 Reader Module to Arduino:** Connect the EM-18's TX pin to one of the Arduino's digital pins configured for serial communication.
- **LCD Display to Arduino:** Connect the LCD using the digital pins on the Arduino (as defined in your code: 8, 9, 10, 11, 12, 13).
- **LEDs to Arduino:** Connect the red and green LEDs to separate digital pins on the Arduino (pins 6 and 7, respectively).
- **Buzzer to Arduino:** Connect the buzzer to a digital pin on the Arduino.
- **Power Supply:** Ensure all components receive the appropriate voltage. The Arduino Uno typically operates at 5V, which can be supplied through its power jack or via USB.

EXAMPLE WORKFLOW:

1. **Student Enters Room:**
 - RFID reader at the entrance detects the tag on the student's ID card.
 - Reader sends the tag ID to the central computing device.
2. **Data Processing:**

CHAPTER 4:

Hardware implementation:

4.1 Arduino uno R3

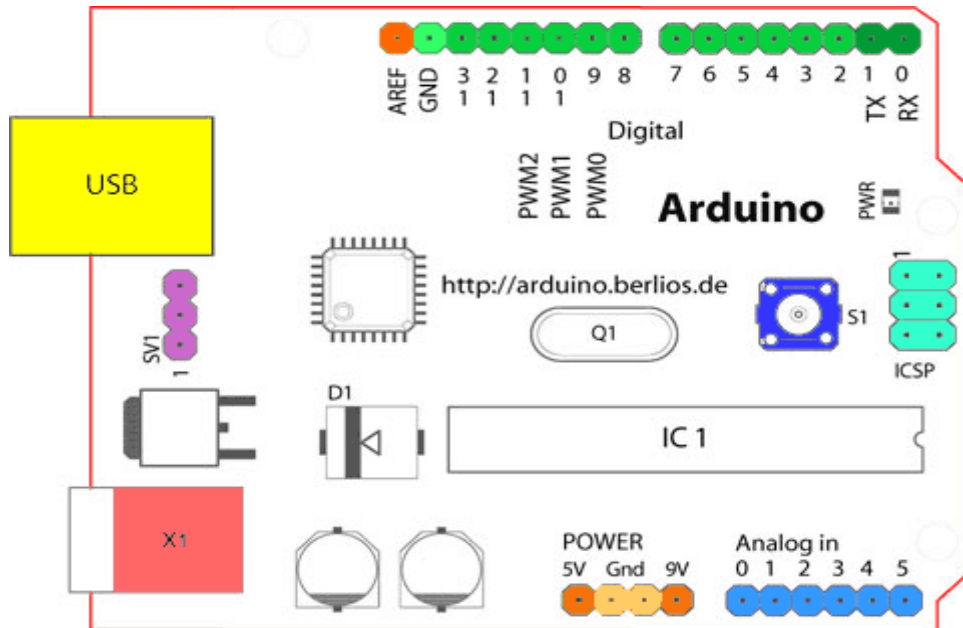


Figure (3)

The Arduino UNO.

Starting clockwise from the top center:

Analog Reference pin (orange)

Digital Ground (light green)

Digital Pins 2-13 (green)

Digital Pins 0-1/Serial In/Out - TX/RX (dark green) - These pins cannot be used for digital i/o (digitalRead and digitalWrite) if you are also using serial communication (e.g. Serial.begin).

Reset Button - S1 (dark blue)

In-circuit Serial Programmer (blue-green)

Analog In Pins 0-5 (light blue)

Power and Ground Pins (power: orange, grounds: light orange)

External Power Supply In (9-12VDC) - X1 (pink)

Toggles External Power and USB Power (place jumper on two pins closest to desired supply)
- SV1 (purple)

USB (used for uploading sketches to the board and for serial communication between the board and the computer; can be used to power the board) (yellow)

4.2 Microcontrollers

ATmega328P (used on most recent boards)

Digital I/O Pins: 14 (of which 6 provide PWM output)

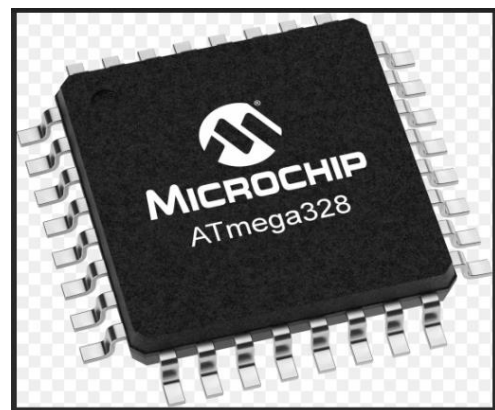
Analog Input Pins: 6 (DIP) or 8 (SMD)

DC Current per I/O Pin: 40 mA

Flash Memory: 32 KB

SRAM: 2 KB

EEPROM: 1 KB



ATMEG328 Microcontroller Figure (4)

ATmega168 (used on most Arduino Diecimila and early Duemilanove)

Digital I/O Pins: 14 (of which 6 provide PWM output)

Analog Input Pins: 6 (DIP) or 8 (SMD)

DC Current per I/O Pin: 40 mA

Flash Memory 16 KB:

SRAM: 1 KB

EEPROM: 512 bytes

ATmega8 (used on some older board)

Digital I/O Pins: 14 (of which 3 provide PWM output)

Analog Input Pins: 6

DC Current per I/O Pin: 40 mA

Flash Memory: 8 KB

SRAM: 1 KB

EEPROM: 512 bytes

Digital Pins

In addition to the specific functions listed below, the digital pins on an Arduino board can be used for general purpose input and output via the `pinMode()`, `digitalRead()`, and `digitalWrite()` commands. Each pin has an internal pull-up resistor which can be turned on and off using `digitalWrite()` (w/ a value of HIGH or LOW, respectively) when the pin is configured as an input. The maximum current per pin is 40 mA.

Serial: 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. On the Arduino Diecimila, these pins are connected to the corresponding pins of the FTDI USB-to-TTL Serial chip. On the Arduino BT, they are connected to the corresponding pins of the WT11 Bluetooth® module. On the Arduino Mini and LilyPad Arduino, they are intended for use with an external TTL serial module (e.g. the Mini-USB Adapter).

External Interrupts: 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the `attachInterrupt()` function for details.

PWM: 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the `analogWrite()` function. On boards with an ATmega8, PWM output is available only on pins 9, 10, and 11.

BT Reset: 7. (Arduino BT-only) Connected to the reset line of the Bluetooth® module.

SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication, which, although provided by the underlying hardware, is not currently included in the Arduino language.

LED: 13. On the Diecimila and LilyPad, there is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

Analog Pins

In addition to the specific functions listed below, the analog input pins support 10-bit analog-to-digital conversion (ADC) using the `analogRead()` function. Most of the analog inputs can also be used as digital pins: analog input 0 as digital pin 14 through analog input 5 as digital pin 19. Analog inputs 6 and 7 (present on the Mini and BT) cannot be used as digital pins.

I2C: 4 (SDA) and 5 (SCL). Support I2C (TWI) communication using the Wire library (documentation on the Wiring website).

Power Pins

VIN (sometimes labelled "9V"). The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin. Note that different boards accept different input voltages ranges, please see the documentation for your board. Also note that the LilyPad has no VIN pin and accepts only a regulated input.

5V. The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via an on-board regulator, or be supplied by USB or another regulated 5V supply.

3V3. (Diecimila-only) A 3.3 volt supply generated by the on-board FTDI chip.

GND. Ground pins.

Other Pins

AREF. Reference voltage for the analog inputs. Used with `analogReference().Reset`. (Diecimila-only) Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

implementing an Arduino Uno hardware project involves planning, connecting the components, writing the code, and testing the system. Here's a step-by-step guide for a basic project, such as controlling an LED with a button.

4.3 LCD Display:

LCD (Liquid Crystal Display) screens are electrical display modules that have a variety of uses. A 20*4 The LCD display is a simple element that can be found in a wide range of devices and circuits. Because of the following reasons, liquid crystal displays are increasingly being used to replace LEDs (seven segment LEDs).

LCDs are becoming less expensive. Numbers, characters, and photos can all be shown. This contrasts with LEDs, which can only display numbers and a few characters.

Inclusion of a smart controller through the LCD, allowing the CPU to handle the process of reloading the liquid crystal display. In contrast, the LED should be reloaded using the CPU to keep records. LCD interfacing with controller:

The LCD protocol includes three control lines and eight I/O lines for the data bus. 8 data pins D7-D0 are Bi-directional data/command pins. Alphanumeric characters are sent in ASCII format.

RS (Register Select) RS = 0 -> command register is selected. RS = 1 -> Data Register is selected. R/W (Read or Write) 0 -> Write, 1 -> Read E: Enable (Latch data) This is used to latch the data on the data pins. To latch the data, a high-to-low edge is required.

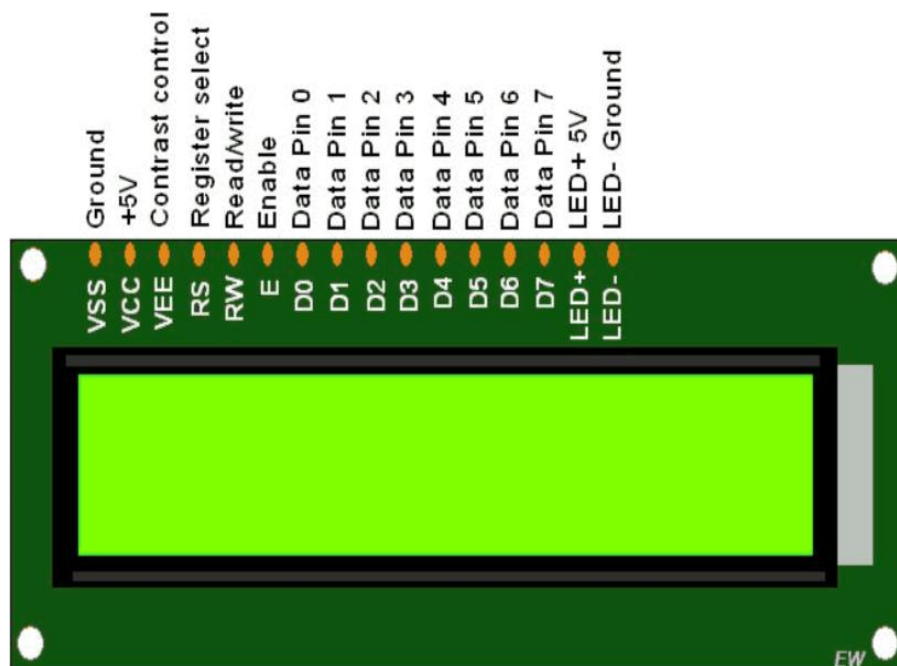


FIG: LIQUID CRYSTAL DISPLAY (LCD)

4.4 RFID Reader:

An RFID reader or scanner operates relatively to a barcode scanner, rather than scanning the barcode with a laser beam, it scans it with electromagnetic waves. The scanner uses an antenna that transmits a signal and interacts with the tag's antenna to send these waves. The antenna on the tag collects input from the scanner and send data about the chip to the scanner.



FIG: EM-18 READER MODULE



FIG: EM-18 READER MODULE PIN CIRCUIT

The data on the chip is generally located in one of two memory types. Read-Only Memory (ROM) is the most prevalent; as the name suggests, read-only memory cannot be modified once it has been encoded onto the chip during the production process. The second type of memory is Read/Write Memory, which can be changed later by certain devices despite being configured during the production line.

4.5 FINGERPRINT SENSOR

- The R307 Optical Fingerprint Reader Sensor is what you're looking at. The R307 fingerprint module has a TTL UART interface for direct connection to a microcontroller UART or a PC through a MAX232. The user can save fingerprint data in the module and set it to identify the person in 1:1 or 1: N mode.

- The FP device can connect directly to a 3.3 or 5 volt microcontroller. Interfacing with a PC serial port necessitates the use of a level converter (such as MAX232).



FIG: FINGERPRINT SENSOR

4.6 BUZZER

A buzzer or beeper is an auditory signalling device that can be mechanical, electromechanical, or piezoelectric. Buzzers and beepers are commonly used for alarm clocks, timers, and confirmation of human input such as a mouse click or keyboard. Buzzers are electronic transducers with a DC power source that are commonly used in sound devices such as computers, printers, alarms, electronic toys, automobile electronic and other electronic products.



FIG: BUZZER

4.7 RFID TAG:

A radio frequency identification tag (RFID tag) is a tiny device that stores and sends data to an RFID reader. There are two sorts of tags: active tags and passive tags. The reader is not required to power active tags because they have an inbuilt battery. The range of active tags is usually larger than that of passive tags. Tags that are active are larger in size than tags that are inactive. They don't have a built-in battery and depend on the RFID reader for power, and they only get a few metres of range.



FIG: RIFD TAG

Advantages of RFID

- Tag identification that is not visible to the naked eye.
- It is feasible to run operations without human intervention, which lowers human error and saves money.
- Capacity to comprehend moving items that have tags attached.
- Wider coverage area, up to a few feet
- Can be employed in a variety of settings, such as cattle, military, and scientific research.
- RFID can be used in conjunction with bar codes. Both the technologies may be mutually beneficial.

CHAPTER 5:

SOFTWARE IMPLEMENTATION:

5.1 ARDUINO UNO SOFTWARE:

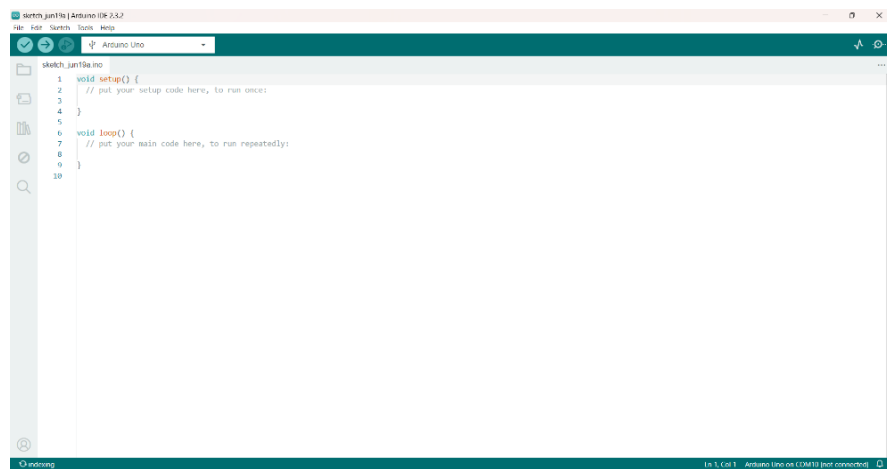
1. DOWNLOAD AND INSTALL THE ARDUINO (IDE)

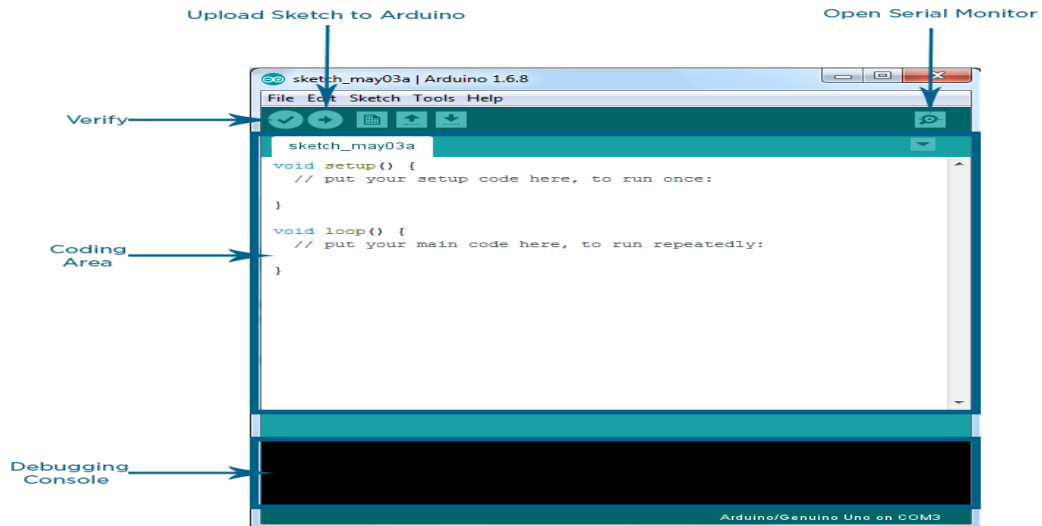
- **Download:** Go to the Arduino Software page and download the appropriate version for your operating system (Windows, macOS, or Linux).
- **Install:**

- **Windows:** Run the downloaded .exe file and follow the installation instructions.
- **macOS:** Open the downloaded .dmg file and drag the Arduino application into the Applications folder.
- **Linux:** Extract the downloaded tarball and run the install.sh script.

2. Launch the Arduino IDE

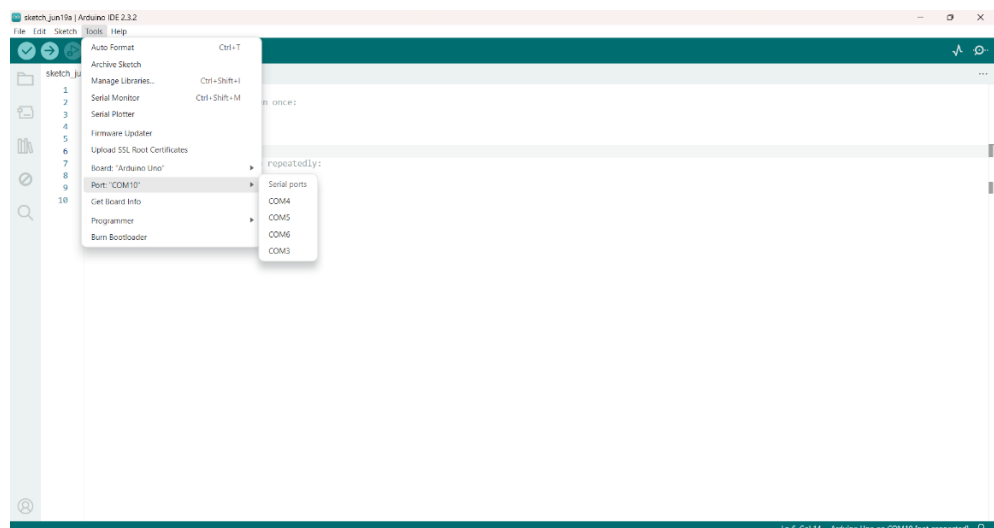
- After installation, open the Arduino IDE. You should see a window with a text editor, a message area, a console, and a toolbar with buttons for common functions.





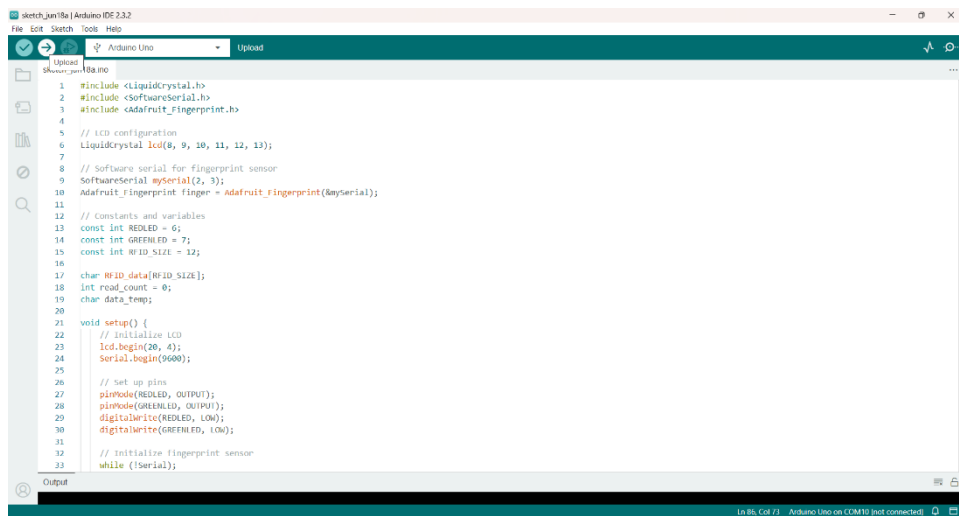
3. Set up your Arduino Uno

- **Connect the Arduino Uno to your computer** using a USB cable.
- **Select the Board:** In the Arduino IDE, go to Tools > Board > Arduino Uno.
- **Select the Port:** Go to Tools > Port and select the port that corresponds to your Arduino Uno (e.g., COM3 on Windows or /dev/tty.usbmodem1411 on macOS).



4. Write and upload code

- **Write a Sketch:** A sketch is the name Arduino uses for a program. Here's a basic example sketch that blinks the onboard LED:
- **Upload the Sketch:** Click the Upload button (right arrow icon) in the Arduino IDE. This compiles the code and uploads it to the Arduino Uno.



5. Using Libraries

- The Arduino IDE comes with many libraries that extend the functionality of your sketches. You can include libraries using Sketch > Include Library.
- To install additional libraries, go to Sketch > Include Library > Manage Libraries and search for the library you need.

Alternative IDEs and Extensions

While the official Arduino IDE is widely used, there are alternative IDEs and extensions that offer additional features:

1.Arduino Web Editor

- An online version of the Arduino IDE that allows you to write, compile, and upload sketches from your web browser. It's accessible from any device with internet access.
- Available at Arduino Web Editor.

2.PlatformIO

- A professional collaborative platform for embedded development that supports Arduino. It integrates with various IDEs like Visual Studio Code.
- More information at [PlatformIO](https://platformio.org/).
-

3.Visual Studio Code with Arduino Extension

- Visual Studio Code (VS Code) with the Arduino extension provides a powerful development environment for Arduino projects.
- Install VS Code from [Visual Studio Code](#).
- Add the Arduino extension from the VS Code marketplace.

Additional Resources

- **Arduino Reference:** Comprehensive documentation on the Arduino programming language and built-in functions. Available at [Arduino Reference](#).
- **Arduino Forum:** A community forum for asking questions and sharing projects. Available at [Arduino Forum](#).
-

5.2 PROGRAM CODE:

```
#include <LiquidCrystal.h>
#include <SoftwareSerial.h>
#include <Adafruit_Fingerprint.h>

// LCD configuration
LiquidCrystal lcd(8, 9, 10, 11, 12, 13);

// Software serial for fingerprint sensor
SoftwareSerial mySerial(2, 3);
Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);

// Constants and variables
const int REDLED = 6;
const int GREENLED = 7;
const int RFID_SIZE = 12;

char RFID_data[RFID_SIZE];
int read_count = 0;
char data_temp;
```

```

void setup() {
    // Initialize LCD
    lcd.begin(20, 4);
    Serial.begin(9600);

    // Set up pins
    pinMode(REDLED, OUTPUT);
    pinMode(GREENLED, OUTPUT);
    digitalWrite(REDLED, LOW);
    digitalWrite(GREENLED, LOW);

    // Initialize fingerprint sensor
    while (!Serial);
    delay(100);
    finger.begin(9600);
    delay(1000);

    lcd.clear();
    lcd.setCursor(0, 0); lcd.print("Examination Room");
    lcd.setCursor(0, 1); lcd.print("Guide System");
    lcd.setCursor(0, 2); lcd.print("Using RFID With");
    lcd.setCursor(0, 3); lcd.print("Fingerprint ");
    delay(6000);
    lcd.clear();

    if (finger.verifyPassword()) {
        lcd.setCursor(0, 0); lcd.print("R307 FINGER FP ");
        delay(1000);
    } else {
        Serial.println("Did not find fingerprint sensor :(");
        while (1) {
            delay(1);
        }
    }
    finger.getTemplateCount();
}

void loop() {
    while (true) {
        handleFingerprint();
    }
}

void handleFingerprint() {
    int id;
    digitalWrite(REDLED, HIGH);
    digitalWrite(GREENLED, LOW);
}

```

```

lcd.setCursor(0, 0); lcd.print("PLACE FINGER FP ");

id = getFingerprintIDez();
if (id >= 0) {
    lcd.setCursor(0, 0); lcd.print("FINGERPRINT SUCCESS");
    lcd.setCursor(0, 1); lcd.print("SHOW RFID CARD ");

    while (true) {
        if (Serial.available() > 0) {
            data_temp = Serial.read();
            RFID_data[read_count] = data_temp;
            read_count++;

            if (read_count == RFID_SIZE) {
                if (!processRFID(id)) {
                    // If RFID does not match, show error message
                    lcd.clear();
                    lcd.setCursor(0, 0); lcd.print("RFID NOT RECOGNIZED");
                    delay(5000);
                    lcd.clear();
                    read_count = 0;
                    break;
                }
                read_count = 0;
            }
        }
    }
}
delay(1000);
}

bool processRFID(int id) {
    lcd.clear();
    for (int i = 0; i < RFID_SIZE; i++) {
        lcd.print(RFID_data[i]);
    }

    if (id == 0 && !strcmp(RFID_data, "150024229182", RFID_SIZE)) {
        displayStudentInfo("G.SUSMITHA", "227R1A0481", "ECE", "32-BLOCK-C");
        return true;
    } else if (id == 1 && !strcmp(RFID_data, "1500271AF9D1", RFID_SIZE)) {
        displayStudentInfo("K.DHANUNJAY", "227R1A0485", "ECE", "32-BLOCK-C");
        return true;
    } else if (id == 2 && !strcmp(RFID_data, "1500270D5B64", RFID_SIZE)) {
        displayStudentInfo("J.DHANUSH", "227R1A0469", "ECE", "33-BLOCK-B");
        return true;
    } else if (id == 3 && !strcmp(RFID_data, "01006EADC103", RFID_SIZE)) {

```

```

        displayStudentInfo("T.KIRAN", "227R1A04B9", "ECE", "32-BLOCK-A");
        return true;
    } else {
        return false;
    }
}

void displayStudentInfo(const char* name, const char* hallTicket, const char*
branch, const char* room) {
    digitalWrite(REDDLED, LOW);
    digitalWrite(GREENLED, HIGH);

    lcd.clear();
    lcd.setCursor(0, 0); lcd.print(name);
    lcd.setCursor(0, 1); lcd.print("HallTicket:"); lcd.print(hallTicket);
    lcd.setCursor(0, 2); lcd.print("BRANCH-"); lcd.print(branch);
    lcd.setCursor(0, 3); lcd.print("ROOM NO:"); lcd.print(room);

    delay(5000);
    lcd.clear();
    digitalWrite(REDDLED, HIGH);
    digitalWrite(GREENLED, LOW);
}

int getFingerprintIDez() {
    uint8_t p = finger.getImage();
    if (p != FINGERPRINT_OK) return -1;

    p = finger.image2Tz();
    if (p != FINGERPRINT_OK) return -1;

    p = finger.fingerFastSearch();
    if (p != FINGERPRINT_OK) return -1;

    return finger.fingerID;
}

```

CHAPTER 6:

CONCLUSION & FUTURE SCOPE:

6.1 CONCLUSION:

We were able to design an embedded system for examination room navigation that was both cost-effective and time-efficient by combining RFID technology and a fingerprint module. The existing method can be improved by adding an infrared sensor and utilised to track attendance. As a result, the system offers a wide range of potential development that can lead to significant advancements.

6.2 FUTURE SCOPE:

In addition to the present system, a facial recognition system can be incorporated for increased safety. More capabilities are being added, such as the ability to keep track of a student's details, such as fees due, library transactions, and attendance. The notion is not only useful to students but also to corporations, depending on how well it is implemented, as seen by the seeds that have sprouted into a variety of viable projects.

CHAPTER 7:

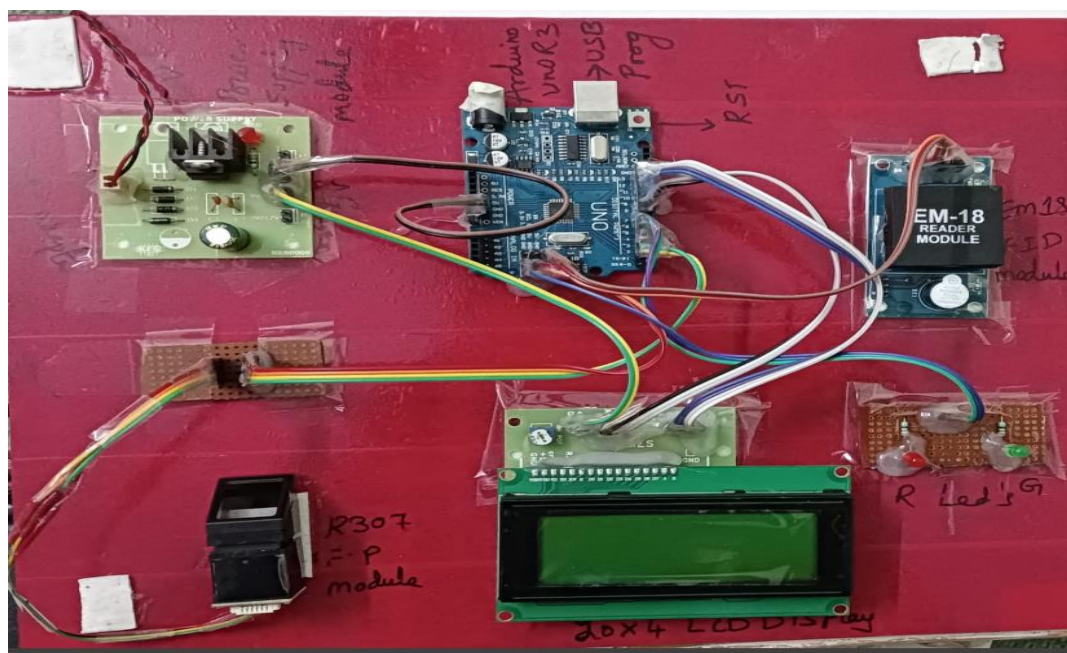
7.1 REFERENCES:

- [1] Smart ambulance technique: Andrea Zanella, Senior Member, IEEE, Nicola Bui, Angelo Castellani, Lorenzo Vangelista, Senior Member, IEEE, and Michele Zorzi, Fellow, IEEE. IEE JOURNAL, VOL. 1, NO. 1, FEBRUARY 2014.
- [2] Veeramuthuvenkatesh, M. prashanthkumar, V.Vaithayanathan, Pethuru Raj, " An ambient healthmonitor for the new generation healthcare," Journal of Theoretical and Applied Information Technology, Vol. 31 No.2, pp. 9199, Sep 2011.
- [3] VeeramuthuVenkatesh, Pethuru Raj, KaushikGopalan and Rajeev.T, " Healthcare Data Fusion and Presentation using Service-Oriented Architecture (SOA) Orchestration Mechanism," IJCA Special Issue on Artificial Intelligence Techniques - Novel Approaches & Practical Applications, Vol. 2, pp. 17-23, June 2011.

- [4] Emergency Management System Using Android Application RehkaJadhav, Jwalant Patel, Darshan Jain, Suyash Phadhtare Department of Information Technology G. H. Raisani Collage of Engineering & Technology, University of Pune,
- [5] Ruha Zhang, and Dongfeng Yuan, "A Health Monitoring System for Wireless Sensor Networks," in Proc. of 2nd IEEE Conference on Industrial Electronics and Applications (ICIEA), pp. 1648-1652, Harbin, China, May 2007.
- [6] A Public Safety Application of GPS-Enabled Smartphones and the Android Operating System, John Whipple William Arensman Marian Starr Boler, Information Systems Engineering Department, Proceedings of the 2009 IEEE International Conference on Systems, Man, and Cybernetics San Antonio, TX, USA - October 2009. SSS
- [7] Grammar-Driven Development of JSON Processing Applications, Antonio Sarasa-Carbazole, José-Luis Sierra, Fac. Informatic. Universidad Completeness de Madrid. 28040 Madrid (Spain), Proceedings of the 2013 Federated Conference on .

CHAPTER 8

8.1 RESULT:



We have tried by using different components.

IMPLEMENTATION 1:

First our project is about RFID AND FINGERPRINT BASED EXAMINATION ROOM IDENTIFICATION. Let us go for the implementation -1. For this first we have to enroll our fingerprints in the fingerprint module by using the SFG software.

1. Say for example let us take person-1 if he keeps his finger on the fingerprint module then we can see “FINGERPRINT SUCCESSFUL SHOW RFID TAG” on the LCD display.



2. Next we need to show the RFID tag, this tag is scanned by the RFID reader module. Then we can observe the first person details (NAME, HALL TICKET, BRANCH, BLOCK, ROOM NO.). At the same time if we observe the green LED light will glow. We can see the output below.



IMPLEMENTATION 2:

Let us take another scenario in this the second person places his/her finger on the finger print module .then the lcd will display that “FINGERPRINT SUCCESSFUL SHOW RFID TAG” so instead of showing the 2nd persons RFID tag we tried with the 1st persons RFID tag .when we place the 1st person RFID tag to the RFID reader module .It will show that “RFID NOT MATCHED WITH THE FINGERPRINT” on the LCD display . At this condition the RED LED will glown . We can see the output below.



IMPLEMENTATION 3:

If we go with the implementation 3 in this we have used ID card (BARCODE) instead of RFID tag . In this case the RFID reader module will not scan the ID card

(BARCODE) of the student. Because RFID reader module will only scan the RFID tag. In this case if we observe the red LED will glow.

IMPLEMENTATION 4:

In this case we have replaced the RFID tag with the QR code (scanner). In this case the RFID reader module will not scan the QR code (scanner). Because RFID reader module will only scan the RFID tag. In this case also the red LED will glow.